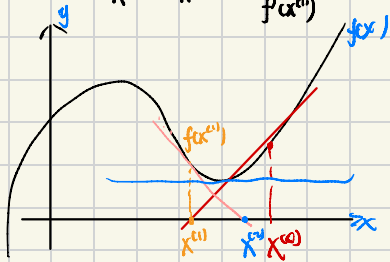
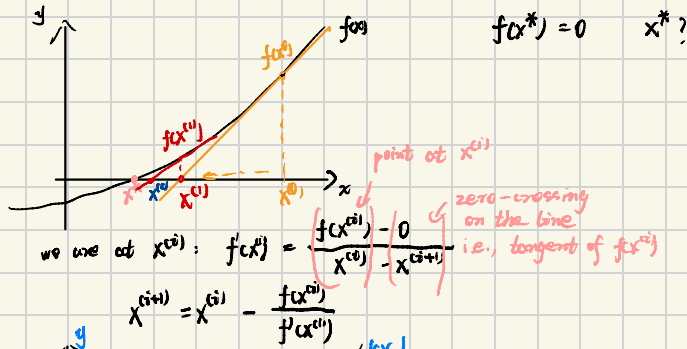


Tutorial 8-6 Newton Raphson Method



find the $\max_x g(x)$

find the zero crossing of $f(x) = g'(x)$

$$x^{(i+1)} = x^{(i)} - \frac{g'(x^{(i)})}{g''(x^{(i)})}$$

$$p(y=1|x) = \sigma(w^T x) = \pi_i \quad \sigma(z) = \frac{1}{1+e^{-z}}$$

$$\text{Train Set: } \{(x_i, y_i)\}_{i=1}^N$$

$$\text{likelihood: } p(y_i=1|x_i) = \pi_i^{y_i} (1-\pi_i)^{(1-y_i)} \\ \pi_i = \sigma(w^T x_i)$$

Add a prior of w

$$p(w) = \mathcal{N}(w|0, \Gamma^{-1})$$

Use MAP estimation:

$$w^* = \arg \max_w p(w|x, y) = \arg \max_w \underbrace{\log p(y|x, w)}_{(1)} + \underbrace{\log p(w)}_{(2)}$$

$$(1) \log p(y|x, w) = \sum_{i=1}^N \log p(y_i|x_i, w) = \sum_{i=1}^N y_i \log \pi_i + (1-y_i) \log (1-\pi_i)$$

$$(2) \log p(w) = -\frac{1}{2} w^T \Gamma w + \text{const.}$$

Find Gradients

$$\frac{\partial}{\partial w} (1) = \sum_{i=1}^N y_i \frac{1}{\pi_i} \frac{\partial \pi_i}{\partial w} + (1-y_i) \frac{1}{1-\pi_i} \frac{\partial (1-\pi_i)}{\partial w}$$

$$\frac{\partial \pi_i}{\partial w} = \frac{\partial}{\partial w} \sigma(w^T x_i) = \sigma'(w^T x_i) \frac{\partial}{\partial w} w^T x_i = \sigma'(w^T x_i) \cdot x_i$$

$$\sigma'(z) = \frac{\partial}{\partial z} \sigma(z) = \frac{\partial}{\partial z} \frac{1}{1+e^{-z}} = \frac{-1}{(1+e^{-z})^2} \cdot \frac{\partial (1+e^{-z})}{\partial z} \\ = \frac{e^{-z}}{(1+e^{-z})^2} \cdot 1 = \sigma(z) \cdot (1-\sigma(z))$$

$$\frac{\partial}{\partial w} (1) = \sum y_i \frac{1}{\pi_i} \pi_i (1-\pi_i) x_i + (1-y_i) \frac{1}{1-\pi_i} \pi_i (1-\pi_i) x_i \\ = \sum y_i x_i - y_i \pi_i x_i - \pi_i x_i + y_i \pi_i x_i \\ = \sum_{i=1}^N (y_i - \pi_i) x_i = X(y - \pi)$$

$$\begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} \pi_1 \\ \vdots \\ \pi_n \end{bmatrix}$$

$$\frac{\partial}{\partial w} (2) = \frac{\partial}{\partial w} -\frac{1}{2} w^T \Gamma w = -\frac{1}{2} \cdot 2 \cdot \Gamma w = -\Gamma w$$

Compute Hessian of (1) & (2)

$$\frac{\partial}{\partial w} \frac{\partial}{\partial w^T} (1) = \frac{\partial}{\partial w} (y - \pi)^T X^T = -\frac{\partial}{\partial w} \pi^T X^T = -X R X^T$$

$$\frac{\partial}{\partial w} \pi^T = \frac{\partial}{\partial w} [\pi_1, \pi_2, \dots, \pi_n] = \begin{bmatrix} \pi_1(1-\pi_1)x_1 & \dots & \pi_n(1-\pi_n)x_n \\ \vdots & & \vdots \end{bmatrix} \\ = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \underbrace{\begin{bmatrix} \pi_1(1-\pi_1) & 0 & \dots & 0 \\ 0 & \pi_2(1-\pi_2) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \pi_n(1-\pi_n) \end{bmatrix}}_R = X R$$

$$\frac{\partial}{\partial w} \frac{\partial}{\partial w^T} (2) = \frac{\partial}{\partial w} (-w^T \Gamma) = -\Gamma$$

$$\frac{\partial}{\partial w} w^T = \begin{bmatrix} \frac{\partial}{\partial w_1} \\ \vdots \\ \frac{\partial}{\partial w_n} \end{bmatrix} [w_1, \dots, w_n] = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 \end{bmatrix} = I$$

Tutorial 8-6 Newton Raphson Method

$$\nabla L = \nabla(\Phi^T \Phi) = X(y - \pi) - \Gamma^T w$$

$$\nabla^2 L = \checkmark = -XRX^T - \Gamma^T$$

$$w^{(i+1)} = w^{(i)} - [\nabla^2 L]^{-1} \nabla L$$

$$= w^{(i)} + (XRX^T + \Gamma^T)^{-1} (X(y - \pi) - \Gamma^T w^{(i)})$$

$$(XRX^T + \Gamma^T)(XRX^T + \Gamma^T)^{-1}$$

$$= (XRX^T + \Gamma^T)^{-1} [XRX^T w^{(i)} + \cancel{\Gamma^T w^{(i)}} + X(y - \pi) - \cancel{\Gamma^T w^{(i)}}]$$

$$= (XRX^T + \Gamma^T)^{-1} \times R \underbrace{[X^T w^{(i)} + R^T (y - \pi)]}_{Z}$$

$$= \underline{(XRX^T + \Gamma^T)^{-1}} \times R Z$$

→ extreme inverse of mat.

last lecture: $(XRX^T)^{-1} \times R Z$